Program 7: Maven Web App

Step 1: Create the project using the command :
“mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false”

```
brijesh@Ubuntu:~/Devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp -DarchetypeArtifactId=mave
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven-
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/brijesh/Devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenWebApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/brijesh/Devops/MyMavenWebApp
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  7.073 s
[INFO] Finished at: 2026-04-08T08:35:11+05:30
```

Step 2: Using tree command to view the structure

```
brijesh@Ubuntu:~/Devops/MyMavenWebApp$ tree
.
├── pom.xml
└── src
    └── main
        ├── resources
        └── webapp
            ├── index.jsp
            └── WEB-INF
                └── web.xml

5 directories, 3 files
```

STEP 3: Update the pom.xml by adding the following dependencies and plugins

```
<!-- Servlet API for Java EE -->
      <dependency>
         <groupId>javax.servlet</groupId>
         <artifactId>javax.servlet-api</artifactId>
         <version>4.0.1</version>
         <scope>provided</scope>
      </dependency>



 <build>
  <plugins>
  <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-war-plugin</artifactId>
    <version>3.4.0</version> <!-- Update to latest stable version -->
</plugin>
</plugins>
    <finalName>MyMavenWebApp</finalName>
  </build>
```

STEP 4: run mvn clean install

```
brijesh@Ubuntu:~/Devops/MyMavenWebApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:MymavenWebApp01 >---------------------
[INFO] Building MymavenWebApp01 Maven Webapp 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/m
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/m
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/m
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/m
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/m
Downloading from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servl
Downloaded from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servle
Downloading from central: https://repo.maven.apache.org/maven2/net/java/jvnet-parent/3/
Downloaded from central: https://repo.maven.apache.org/maven2/net/java/jvnet-parent/3/jv
Downloading from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servl
Downloaded from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servle
```

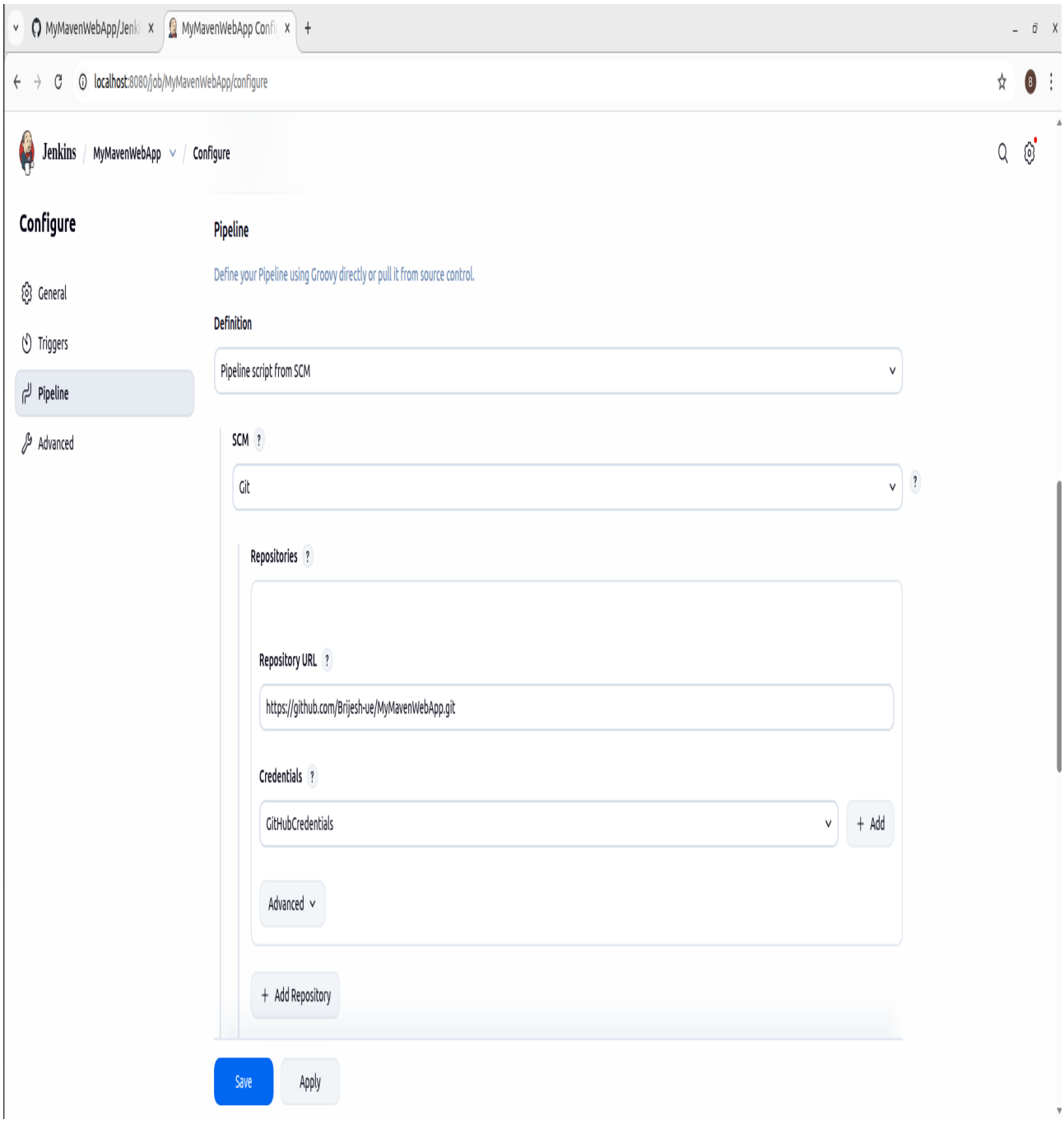
STEP 5: Verify tree structure

```
brijesh@Ubuntu:~/Devops/MyMavenWebApp$ tree
.
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── index.jsp
│           └── WEB-INF
│               └── web.xml
└── target
    ├── classes
    ├── maven-archiver
    │   └── pom.properties
    ├── MymavenWebApp01
    │   ├── index.jsp
    │   ├── META-INF
    │   └── WEB-INF
    │       ├── classes
    │       └── web.xml
    └── MymavenWebApp01.war
```

MyMavenWebApp pipeline:

Step 1: Create a New Pipeline Job
Go to Dashboard → + New Item.
Enter name, select Pipeline, and click OK.

Step 2:Configure Pipeline from SCM
Scroll down and select Pipeline script from SCM.
SCM: Git Enter repository URL and select GitHub credentials.

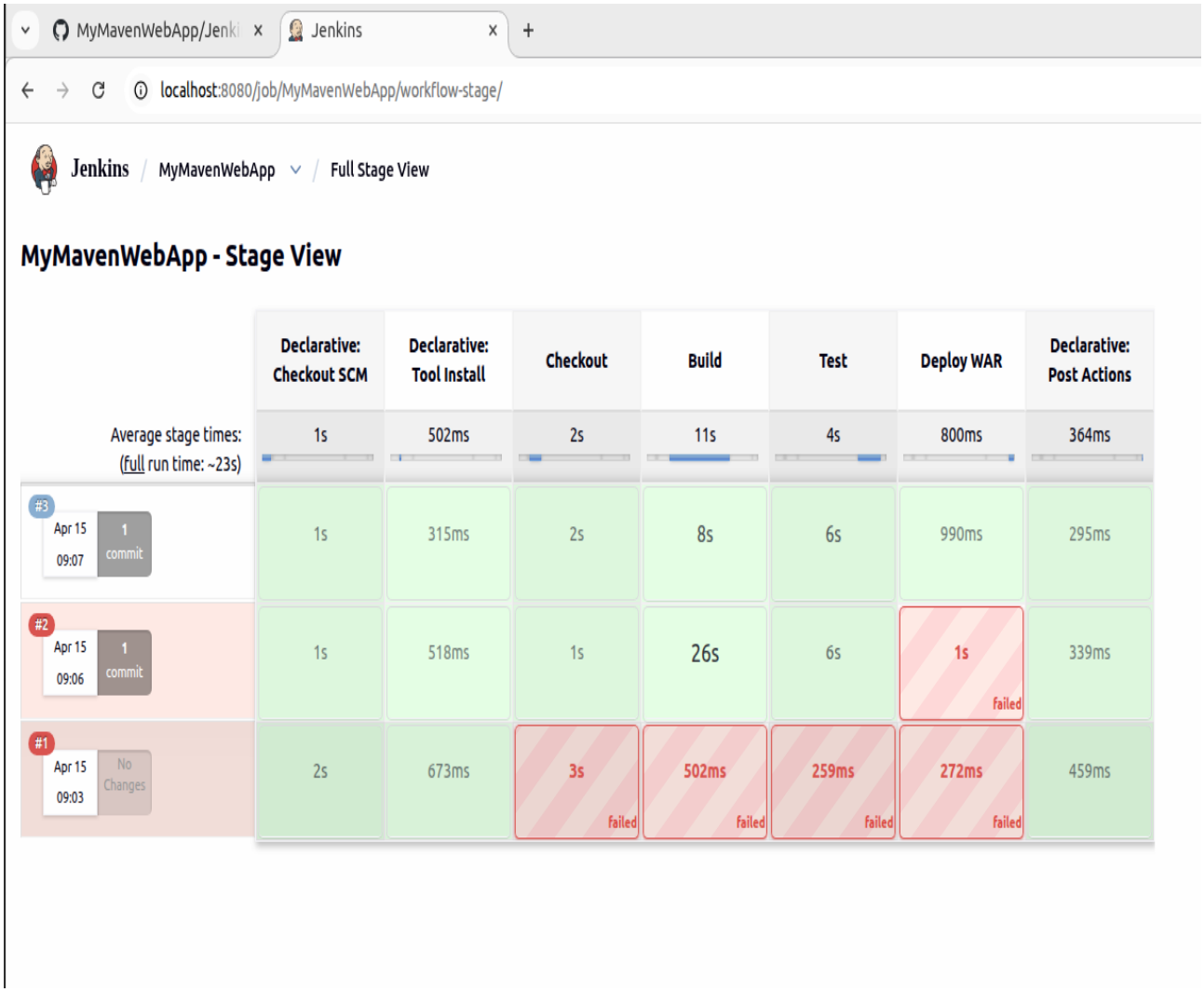


Step 3: Create Jenkinsfile

```
pipeline {
    agent any  // Use any available agent

    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url:
'https://github.com/Brijesh-ue/MyMavenWebApp.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }




        stage('Deploy WAR') {
            steps {
                sh 'cp target/MymavenWebApp01.war /opt/tomcat/webapps/'
            }
        }


    }
```

```
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 4: Build the Pipeline

Go to Jenkins and click Build Now.

Monitor Build Stages Use Stage View and Full Stage View to monitor execution